

MCCREARY, MB, Canada

WMO: 711490

Lat: 50.7119N

Lon: 99.5300W

Elev: 1152

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-16.6	-28.4	1.2	-19.8	-23.9	1.5	-15.5	27.2	32.1	24.2	27.1	5.2	320	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	85.6	68.8	82.1	67.1	79.3	65.3	72.3	80.9	70.0	78.8	67.9	76.4	8.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	113.6	77.3	66.9	103.4	74.8	64.6	95.3	72.8	36.8	81.0	34.7	78.9	32.9	76.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.6	16.3	DB	-28.6	92.3	4.6	3.5	-31.9	94.9	-34.5	97.0	-37.1	98.9	-40.4	101.5	(4)
(5)			WB	-28.8	77.1	4.5	3.5	-32.1	79.6	-34.7	81.6	-37.2	83.6	-40.5	86.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	7.4	10.0	22.5	39.2	51.0	60.8	67.1	65.0	55.7	41.6	25.9	11.4		
	DBStd	23.94	15.29	13.88	14.02	11.46	9.21	6.92	5.99	6.36	8.27	9.24	12.41	14.48		
	HDD50	6000	1319	1120	852	352	99	6	0	0	37	293	725	1197		
	HDD65	9996	1784	1540	1316	773	441	159	48	80	295	726	1172	1662		
	CDD50	1730	0	0	0	30	132	330	530	465	208	32	3	0		
	CDD65	254	0	0	0	1	9	33	114	80	16	1	0	0		
	CDH74	2292	0	0	0	9	127	288	973	716	167	12	0	0		
	CDH80	543	0	0	0	0	23	52	238	198	31	1	0	0		
	WSAvg	7.6	7.8	7.7	8.0	8.4	8.3	6.8	6.7	6.8	7.4	8.3	8.0	7.5		
	PrecAvg	21.9	0.8	0.6	1.3	1.3	2.9	3.7	3.0	2.5	2.1	1.6	0.9	0.9		
(15) Precipitation	PrecMax	26.7	2.0	2.1	4.4	2.4	7.3	6.5	6.9	6.4	3.9	5.1	4.1	2.2		
	PrecMin	16.0	0.1	0.0	0.0	0.3	0.8	1.4	1.0	0.4	0.4	0.2	0.0	0.1		
	PrecStd	2.9	0.5	0.4	1.0	0.7	1.5	1.4	1.5	1.5	1.0	1.1	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	45.4	58.3	74.5	82.6	85.7	89.6	90.5	84.0	74.5	61.4	46.4		
		MCWB	38.5	37.2	45.8	54.7	60.2	68.4	73.0	70.5	64.9	57.3	48.8	37.9		
	2%	DB	40.1	39.5	51.7	68.2	77.6	80.5	85.7	85.2	78.8	66.2	53.3	41.1		
		MCWB	34.9	33.8	42.6	50.9	57.2	65.5	70.4	68.1	64.5	52.7	43.4	35.0		
	5%	DB	35.4	35.2	46.3	63.1	72.6	77.2	82.4	81.3	74.3	61.0	47.4	36.2		
		MCWB	31.2	31.8	39.3	48.8	55.5	63.2	68.6	67.1	61.5	50.0	39.7	31.6		
	10%	DB	30.3	31.1	41.4	57.7	67.5	73.9	79.5	77.7	69.8	56.3	42.5	31.1		
		MCWB	27.5	28.7	36.5	46.0	53.2	61.9	67.0	65.3	58.6	47.7	36.9	27.9		
	0.4%	WB	38.5	37.8	47.4	56.3	64.3	73.6	77.4	74.6	70.2	58.2	49.0	38.3		
		MCDB	44.9	43.0	54.8	69.2	76.3	80.4	84.7	84.5	77.8	70.6	61.0	45.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.2	34.8	43.1	52.3	60.2	68.7	73.2	71.2	66.5	54.1	44.2	35.2		
		MCDB	40.0	38.7	50.8	65.4	70.6	76.4	81.5	80.7	74.8	63.4	52.0	40.7		
	5%	WB	31.5	31.9	39.6	49.5	57.5	65.7	70.7	69.1	63.2	51.2	40.4	31.9		
		MCDB	34.8	35.2	45.6	60.9	68.6	73.1	79.4	78.4	71.2	58.7	46.4	36.0		
	10%	WB	27.5	28.4	36.5	46.7	54.9	63.4	68.6	66.8	60.3	48.5	37.2	28.2		
		MCDB	30.3	31.2	41.3	57.0	64.9	71.4	77.1	75.4	68.1	55.5	42.4	30.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	16.9	17.5	19.9	20.6	18.3	19.2	19.7	18.9	16.1	14.2	15.5		
		MCDBR	20.4	19.6	21.3	27.7	27.8	22.9	22.8	25.0	25.9	24.6	20.7	19.6		
		MCWBR	16.1	15.2	14.2	15.2	13.8	12.2	12.1	13.1	14.1	13.8	13.1	15.2		
		MCDBR	19.6	18.3	20.4	26.0	23.4	20.3	21.4	21.8	23.4	22.5	20.0	19.3		
	5% WB	MCWBR	16.0	14.2	14.0	15.0	13.5	12.2	12.7	13.0	14.7	13.4	13.5	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.259	0.297	0.342	0.357	0.370	0.385	0.378	0.342	0.295	0.263	0.237		
	taud		2.288	2.298	2.316	2.369	2.391	2.379	2.359	2.375	2.459	2.553	2.457	2.353		
	Ebn at Noon		250	282	288	286	285	281	274	271	270	268	249	243		
	Edh at Noon		21	27	32	35	36	36	37	34	28	21	18	16		
(44) All-Sky Solar Radiation	RadAvg		417	795	1240	1616	1760	1852	1958	1624	1167	683	368	297		
	RadStd		34	48	92	110	159	130	86	123	94	88	51	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-1.87	-2.74	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	-0.90	-1.21	N/A	N/A	N/A	N/A

Nomenclature: See separate page